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An Internship Report on

Education App using AI for Zoodle Me

Submitted partial fulfillment of the requirements for the degree of

BACHELOR OF TECHNOLOGY

in

Artificial Intelligence and Data Science

By

S Praveen Venkata Subramani (2362150)

Under the Guidance of

Babu Kumar

Department of AI and Data Science Engineering
School of Engineering and Technology,
CHRIST (Deemed to be University),
Kumbalagodu, Bengaluru - 560 074

July – 2026



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“To excel in human-centred AI and data-driven innovation through ethical research, societal well-being, and transformative collaborations. To excel in human-centred AI and data-driven innovation through ethical research, societal well-being, and transformative collaborations.”

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- M3:** *Innovate scientific knowledge and Entrepreneurship through academia and Industry collaborations.*

Program Educational Objectives (PEOs)

- PEO1:** *Professional Acumen: Understand, analyze and design solutions with professional competency for the real-world problems.*
- PEO2:** *Critical Analysis: Develop software/embedded solutions for the requirements, based on critical analysis and research.*
- PEO3:** *Team work: Function effectively in a team and as an individual in a multidisciplinary/multicultural environment*
- PEO4:** *Life Long Learning: Accomplish holistic development comprehending professional responsibilities.*

Program Outcomes (POs)

PO1: Engineering Knowledge *Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization to develop the solution of complex engineering problems.*

PO2: Problem Analysis *Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.*

PO3: Design/Development of Solutions *Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.*

PO4: Conduct Investigations of Complex Problems *Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.*

PO5: Engineering Tool Usage *Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.*

PO6: The Engineer and The World *Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.*

PO7: Ethics *Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.*

PO8: Individual and Collaborative Team work *Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.*

PO9: Communication *Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.*

PO10: Project Management and Finance *Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.*

PO11: Life-Long Learning *Recognize the need for, and have the preparation and ability for (i) independent and life-long learning (ii) adaptability to new and emerging technologies and (iii) critical thinking in the broadest context of technological change.*

Program Specific Outcomes (PSOs)

PSO1: *Artificial Intelligence: To utilize artificial intelligence principles to solve real world problems.*

PSO2: *Machine Learning To select suitable machine learning algorithms for the given problem.*

PSO3: *Service Learning Analyze Social Relevant Problems and design solutions through Service Learning.*



School of Engineering and Technology

Department of AI and Data Science Engineering

CERTIFICATE

This is to certify that **S Praveen Venkata Subramani (2362150)** has successfully completed his/her internship work entitled “**Education App using AI for Zoodle Me**” at **Cognivo (OPC) Private Limited (Authorized Developer of ZoodleMe™), Uttar Pradesh, India (Remote)**, from **02 April 2026** to **03 May 2026** for a duration of **32** days and submitted on 01 July 2026 in partial fulfillment for the award of **Bachelor of Technology** in **Artificial Intelligence and Data Science** during the academic year **2026–27**.

GUIDE

Babu Kumar

(Signature with Date)

HEAD OF THE DEPARTMENT

Michael Moses T

(Signature with Seal)

EXAMINER 1

(Name & Signature with Date)

Examiner 2

(Name & Signature with Date)

ACKNOWLEDGEMENT

We would like to thank Dr Rev Fr Joseph C. C., Vice Chancellor, CHRIST (Deemed to be University), Dr Rev Fr Viju P. D., Pro Vice Chancellor, Dr Fr Jiby Jose, Director, School of Engineering and Technology, Fr Shijin P. J., Assistant Director, School of Engineering and Technology, CHRIST (Deemed to be University), Dr Raghunandan Kumar R., Dean, and Dr Mary Anitha E. A., Associate Dean, for their kind patronage.

We would also like to express our sincere gratitude and appreciation to **Michael Moses T**, Head of the Department, **Department of AI and Data Science Engineering**, for giving us this opportunity to take up this internship and for the constant encouragement extended throughout the programme.

We are extremely grateful to our faculty guide, **Babu Kumar**, who has supported and helped us carry out this internship work. His constant monitoring, timely guidance, and encouragement helped us keep up to the internship schedule.

We are also extremely grateful to our industry guide, **Anuj Yadav**, Chief Executive Officer, **Cognivo (OPC) Private Limited**, the authorized developer of the ZoodleMe™ platform, for entrusting us with meaningful responsibilities, sharing valuable technical insights, and mentoring us through the design and development of the AI, machine learning, and backend modules of the ZoodleMe educational application.

Finally, we thank all the faculty and non-teaching staff of the department, and our family and friends, for their continuous support and motivation, without which this internship would not have been possible.

DECLARATION

We hereby declare that the project titled “**Education App using AI for Zoodle Me**” is a record of the original internship work undertaken as part of the course **CSE785 – Internship - II**, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology**.

We completed this internship under the supervision of **Babu Kumar** and **Anuj Yadav**, at **Cognivo (OPC) Private Limited (Authorized Developer of ZoodleMe™), Uttar Pradesh, India (Remote)**, during the period from **02 April 2026** to **03 May 2026**.

We further declare that this internship report has not been submitted, either in whole or in part, for the award of any degree, diploma, associateship, fellowship, or any other academic qualification at any other institution. Furthermore, this report has not been submitted for publication or presentation elsewhere.

We affirm that this internship work has been carried out in accordance with the Code of Research Conduct prescribed by the University.

Place: CHRIST (Deemed to be University), Kumbalagodu, Bengaluru - 560 074

Date: 01 July 2026

Signature: S Praveen Venkata Subramani (2362150)

ABSTRACT

This internship report presents the design and development of **ZoodleMe**, an Artificial Intelligence powered educational application built to help students learn faster by converting their own study material into structured, exam-ready learning resources. The internship was carried out at Cognivo (OPC) Private Limited, the authorized developer of the ZoodleMe platform, in the role of AI, Machine Learning, and Backend Developer. The core problem addressed by the project is the significant time and effort students spend manually condensing lecture notes, textbooks, and handwritten material into summaries, flashcards, and practice questions.

The solution allows a learner to upload a photograph of their notes or paste raw text and, within seconds, receive an automatically generated summary, a set of key concepts, interactive flashcards, multiple-choice quizzes, and a glossary of important terms. This is achieved through a pipeline that combines Optical Character Recognition for image-to-text conversion with Large Language Model based content generation, exposed to the mobile and web clients through a secure backend REST API. A gamification layer tracks experience points, streaks, badges, and per-subject mastery to sustain learner engagement. The report documents the system architecture, the backend and AI service implementation, the challenges encountered, and the outcomes achieved during the internship period.

Keywords: Artificial Intelligence, Educational Technology, Large Language Models, Optical Character Recognition, Backend Development, Mobile Application

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CHAPTER 1

INTRODUCTION

The rapid growth of digital learning has changed the way students prepare for examinations and engage with academic content. Learners today have access to an enormous volume of material in the form of lecture slides, textbooks, handwritten notes, and online resources. However, the effort required to convert this raw material into concise and revisable study aids remains almost entirely manual. Students spend a considerable amount of time summarising chapters, preparing flashcards, and creating practice questions, which reduces the time available for actual learning and revision.

ZoodleMe is an Artificial Intelligence powered educational application designed to address this problem. The application allows a student to upload a photograph of their notes or paste text directly, and then automatically generates a structured summary, a list of key concepts, interactive flashcards, multiple-choice quizzes, and a glossary of important terms. By automating the most repetitive part of exam preparation, the application enables the learner to focus on understanding and practice rather than manual note-processing.

This internship was undertaken at **Cognivo (OPC) Private Limited**, the authorized developer of the ZoodleMe™ platform, in the role of AI, Machine Learning, and Backend Developer. The work involved contributing to the artificial intelligence and backend components that power the learning experience, as well as taking ownership of the development and maintenance of the ZoodleMe website. This report describes the objectives of the internship, the profile of the host organisation, the technical implementation of the project, and the outcomes achieved.

1.1 Objective

The primary objectives of the internship were as follows:

- To contribute to the design and development of the AI, machine learning, and backend components of the ZoodleMe educational application.
- To implement an automated content-generation pipeline that converts uploaded notes into summaries, flashcards, quizzes, and key terms.
- To integrate Optical Character Recognition so that image-based notes can be processed into machine-readable text.
- To design and maintain secure backend REST APIs that connect the mobile and web clients to the AI services and the database.
- To implement a gamification layer that tracks experience points, streaks, badges, and per-subject mastery to improve learner engagement.

- To develop and maintain the ZoodleMe website in coordination with the product team.

1.2 Company Profile

Cognivo (OPC) Private Limited is a technology company and the authorized developer of the ZoodleMe™ platform, a premier educational initiative of Adhiti Consulting Pty Ltd, Sydney, Australia. The company focuses on building AI-driven products for the education sector, with the goal of making learning smarter, more personalised, and more accessible for young minds.

The organisation operates a remote-first engineering culture in which interns work on assigned tasks under the guidance of a project lead, participate in team meetings and design discussions, and are responsible for the timely delivery of quality work. During this internship, the work was carried out remotely under the mentorship of the Chief Executive Officer, Mr Anuj Yadav. All work products, code, and documentation generated during the internship are the intellectual property of Cognivo (OPC) Private Limited and its parent entity, Adhiti Consulting Pty Ltd.

CHAPTER 2

TECHNICAL DESCRIPTION AND IMPLEMENTATION

2.1 Actual Work

The internship work centred on building the artificial intelligence and backend layers that power the ZoodleMe learning experience, and on maintaining the ZoodleMe website. This chapter describes the prerequisites, the system architecture, a representative part of the implementation, the responsibilities handled, and the challenges encountered during the internship.

2.1.1 Prerequisites

The following knowledge and tools were required to carry out the assigned work:

- Proficiency in a backend programming language and the Node.js runtime.
- Understanding of REST API design, JSON, and token-based authentication.
- Familiarity with Large Language Model (LLM) prompting and Optical Character Recognition (OCR).
- Working knowledge of a NoSQL database (MongoDB) for storing users, notes, and generated content.
- Basic front-end skills (React / React Native) for integrating and maintaining the website.
- Version control using Git and collaborative development practices.

2.1.2 System Architecture

The ZoodleMe platform follows a layered client–server architecture. The mobile application and the website form the presentation layer and communicate with a backend REST API. The API delegates content generation to an AI service layer that combines OCR with LLM-based generation, and it persists all data in a database. A gamification engine updates the learner’s progress after each activity. The overall flow of data through these layers is shown in Figure 2.1.

When a learner uploads an image of their notes, the backend first passes it to the OCR engine to extract text. The extracted text (or text pasted directly by the learner) is then sent to the LLM generation service, which returns a structured response containing a summary, key concepts, flashcards, quiz questions, and key terms. The result is stored in the database and returned to the client for display.

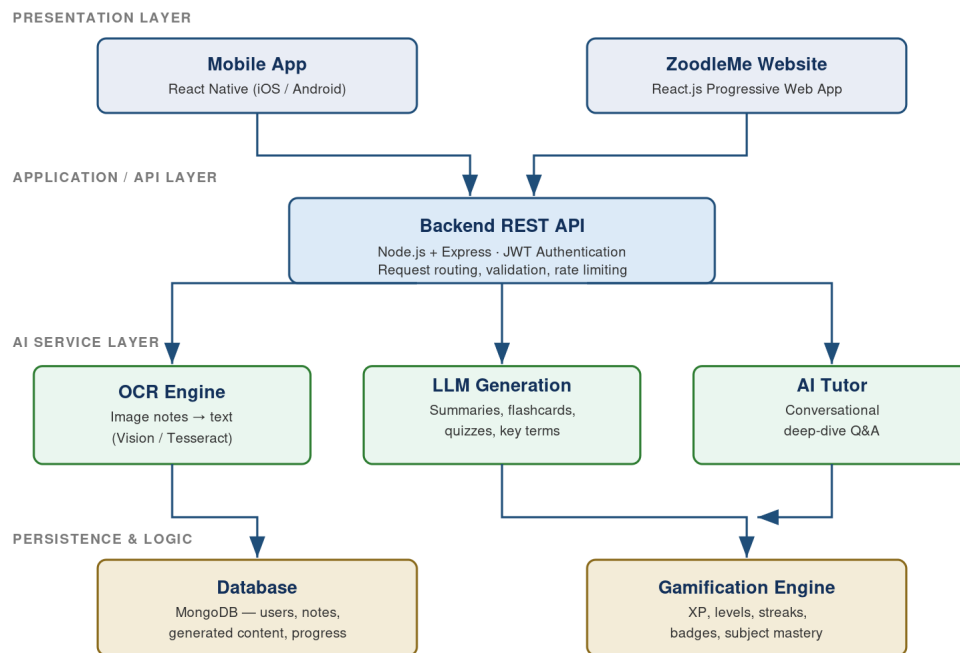


Figure 2.1: System architecture of the ZoodleMe educational application

2.1.3 Implementation

The backend was implemented in Node.js with the Express framework. A single content-generation endpoint orchestrates OCR, LLM generation, and persistence. A representative and simplified version of this endpoint is shown in Listing 2.1. Only the essential logic is shown; error handling, validation, and authentication middleware have been omitted for brevity.

```

1 // POST /api/notes/generate
2 router.post("/generate", auth, async (req, res) => {
3   const { subject, text, imageUrl } = req.body;
4
5   // 1. Obtain the study text (OCR for images, or raw text)
6   const source = imageUrl
7     ? await ocr.extractText(imageUrl)
8     : text;
9
10  // 2. Generate structured study material with the LLM
11  const material = await llm.generate({
12    subject,
13    source,
14    outputs: ["summary", "concepts", "flashcards", "quiz",
15             "terms"]
16  });

```

```

17 // 3. Persist and update learner progress
18 const note = await Note.create({ user: req.user.id, subject,
19   material });
20 await gamification.award(req.user.id, "note_generated");
21 return res.status(201).json(note);
22 });

```

Listing 2.1: Simplified content-generation endpoint

The LLM service builds a carefully structured prompt so that the model returns a predictable JSON object, which is validated before being stored. The upload, summary, and quiz screens produced by this pipeline are shown in Appendix A.3.

2.1.4 Responsibilities

The key responsibilities handled during the internship were:

- Developing and testing backend REST APIs for note upload, content generation, and progress tracking.
- Integrating the OCR and LLM services into the content-generation pipeline.
- Designing the database schema for users, notes, generated material, and gamification data.
- Developing and maintaining the ZoodleMe website.
- Participating in team meetings and design discussions, and delivering assigned tasks on schedule.

2.1.5 Learner Mastery Score

To drive the per-subject mastery indicator shown on the profile screen, a weighted score is computed for each subject from the learner’s quiz accuracy and the number of flashcards reviewed. The mastery score M for a subject is defined as

$$M = 100 \times \left(\alpha a + (1 - \alpha) \frac{r}{r + k} \right), \quad (2.1)$$

where a is the average quiz accuracy in the subject, r is the number of flashcards reviewed, k is a smoothing constant, and $\alpha \in [0, 1]$ balances the two factors.

2.1.6 Flashcard Scheduling Algorithm

Flashcards are presented using a lightweight spaced-repetition schedule so that cards the learner finds difficult reappear sooner. The scheduling logic is summarised in Algorithm 1.

Require: card c with ease factor e_c and interval I_c ; recall quality $q \in \{0, \dots, 5\}$
Ensure: updated interval I_c and next review date

```

1: if  $q < 3$  then
2:    $I_c \leftarrow 1$  // poor recall: review again next day
3: else if  $c$  is new then
4:    $I_c \leftarrow 1$ 
5: else
6:    $I_c \leftarrow \lceil I_c \times e_c \rceil$  // grow the interval by the ease factor
7: end if
8:  $e_c \leftarrow \max(1.3, e_c + 0.1 - (5 - q) \times 0.08)$ 
9: next review date  $\leftarrow$  today  $+ I_c$  days

```

Algorithm 1: Spaced-repetition flashcard scheduling

2.1.7 Module Summary

Table 2.1 summarises the main modules delivered during the internship and their status at the end of the engagement.

Table 2.1: Summary of modules delivered during the internship

Module	Primary Technology	Status
Authentication API	Node.js, JWT	Completed
Note upload & OCR	Express, OCR service	Completed
AI content generation	LLM service	Completed
Gamification engine	Node.js, MongoDB	Completed
ZoodleMe website	React	Maintained

2.1.8 Challenges

Several challenges were encountered during the internship. Ensuring that the LLM consistently returned valid, well-structured output required careful prompt design and strict validation of the model response. Extracting clean text from photographs of handwritten or low-quality notes through OCR was error-prone and needed pre-processing. Finally, keeping API response times low while performing OCR and LLM generation demanded asynchronous processing and sensible request handling. Working through these challenges provided valuable practical experience in building reliable AI-backed systems.

CHAPTER 3

CONCLUSION AND FUTURE SCOPE

3.1 Conclusion

This internship provided valuable hands-on experience in building an Artificial Intelligence powered educational product from concept to working implementation. Working as an AI, Machine Learning, and Backend Developer on the ZoodleMe application, we contributed to the content-generation pipeline that converts a student's notes into summaries, flashcards, quizzes, and key terms, and to the backend REST APIs and gamification logic that support the learning experience. We also took ownership of the development and maintenance of the ZoodleMe website. The internship strengthened our understanding of REST API design, database modelling, LLM prompting, OCR integration, and the practical engineering required to make these components work together reliably. Beyond the technical skills, the experience improved our ability to work within a remote team, communicate progress, and deliver assigned tasks on schedule.

3.2 Future Scope

The ZoodleMe platform can be extended in several directions:

- **Personalised learning paths** that adapt content difficulty to each learner's performance.
- **Support for more content formats**, such as PDFs, audio lectures, and video transcripts.
- **Improved OCR** for handwritten and multilingual notes to broaden accessibility.
- **Offline mode** so that generated material and flashcards can be reviewed without connectivity.
- **Richer analytics** for students and educators to track progress over time.
- **Collaborative study groups** that allow learners to share generated material and compete on leaderboards.

APPENDICES

A.1 Offer Letter



Cognivo (OPC) Private Limited.
(An Authorized Developer of ZoodleMe™)
+918792345338 | admin@zoodleme.com
CIN: U90009UP2025OPC218675

Date: 21st March 2026

To,
S Praveen Venkata Subramani
Hosur, 635126
6383878022
praveensuresh89881@gmail.com

Subject: Internship Offer for ZoodleMe Project

Dear Praveen,

We are delighted to extend to you an offer for an **Internship** at **Cognivo (OPC) Private Limited**, working on the exciting **ZoodleMe** project, a premier initiative of **Adhiti Consulting Pty Ltd, Sydney, Australia**.

This letter outlines the key terms and conditions of your internship:

1. Internship Details

- **Designation:** AI, Machine Learning, and Backend Developer
- **Project:** ZoodleMe – AI-powered Educational App
- **Internship Period:** From 2nd April 2026 to 3rd May 2026 (subject to extension based on performance)
- **Work Mode:** Remote

2. Stipend

- **Compensation:** This is an unpaid, non-stipendiary position. No monetary compensation, stipend, or allowance will be provided.
- No reimbursement will be made for travel, internet, or any other expenses incurred in connection with this role, unless expressly agreed upon in writing.

3. Duties and Responsibilities

You will work on assigned tasks under the guidance of your mentor or project lead. Responsibilities include but are not limited to:

- Assisting in backend/frontend/app development.
- Participating in team meetings and design discussions.
- Timely delivery of assigned work with quality and commitment.

4. Confidentiality & Intellectual Property

During and after your internship, you are required to:

Maintain confidentiality of all proprietary information related to ZoodleMe and its parent entity, Adhiti Consulting Pty Ltd.

- Not share, publish, or disclose any project-related information without written consent.
- Acknowledge that all work products, code, documentation, and ideas generated during your internship are the sole intellectual property of Cognivo (OPC) Private Limited and Adhiti Consulting Pty Ltd.

5. Termination

This internship may be terminated by either party with a notice period of 7 days. Immediate termination without notice may occur in case of misconduct, breach of confidentiality, or failure to comply with company policies.

6. Completion Certificate

Upon satisfactory completion, you will be awarded an Internship Completion Certificate, and based on business needs, an offer for future roles may be considered.

7. Acceptance

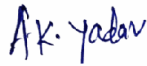
Please confirm your acceptance by signing and returning a copy of this letter.

We are thrilled to have you join us on this inspiring journey to revolutionize education for young minds!

Welcome to the ZoodleMe family!

For Cognivo (OPC) Private Limited

Anuj Yadav
CEO



Acceptance of Offer

I, **S Praveen Venkata Subramani**, accept the terms and conditions outlined in this Internship Offer Letter.

Signature: S Praveen Venkata Subramani

Date: 22nd March, 2026

A.2 Completion Certificate

Cognivo (OPC) Private Limited
An Authorized Developer of ZoodleMe™

+91 90002 64871
admin@zoodleme.com
CIN: U90009UP2025OPC218675

Date: 17 May 2026

INTERNSHIP COMPLETION CERTIFICATE

TO WHOM IT MAY CONCERN

This is to certify that Mr. S Praveen Venkata Subramani has successfully completed his internship with Cognivo (OPC) Private Limited, an authorized developer of the ZoodleMe™ platform (a registered trademark of Adhiti Consulting Pty Ltd, Sydney, Australia).

Internship Details

Name: Mr. S Praveen Venkata Subramani

Internship Duration: 2 April 2026 to 3 May 2026 (1 month)

Designation: AI, Machine Learning and Backend Developer

Project: ZoodleMe - AI-powered Educational Application

Work Mode: Remote

During the course of his internship, Praveen contributed to the design and development of the AI, ML, and backend development components that power the learning experience on the ZoodleMe platform, and took ownership of the development and ongoing maintenance of the ZoodleMe website. He discharged his responsibilities with technical competence, commitment, and professionalism throughout the engagement.

His conduct during the internship was found to be excellent in every respect. We wish him continued success in his future academic and professional endeavors.

For Cognivo (OPC) Private Limited



Anuj Yadav
Chief Executive Officer

COGNIVO (OPC) PRIVATE LIMITED

DIRECTOR

ZoodleMe™ is a registered trademark of Adhiti Consulting Pty Ltd, Sydney, Australia. Cognivo (OPC) Pvt Ltd is an authorized entity developing ZoodleMe projects in India.

A.3 Code and Screenshots

Selected screens of the ZoodleMe educational application produced during the internship are shown below.

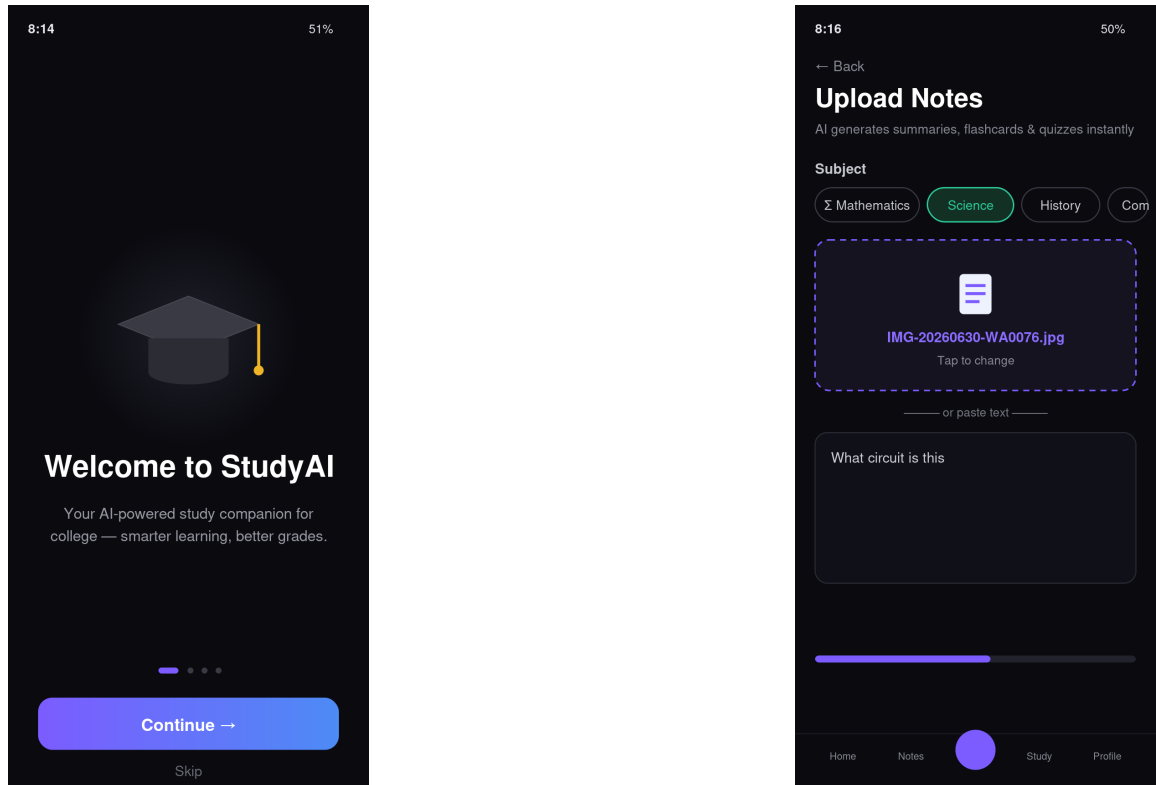


Figure A.1: Onboarding screen (left) and the note-upload screen (right)

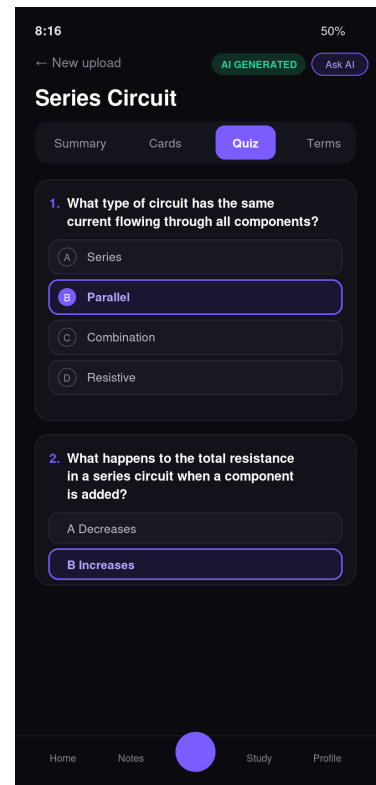
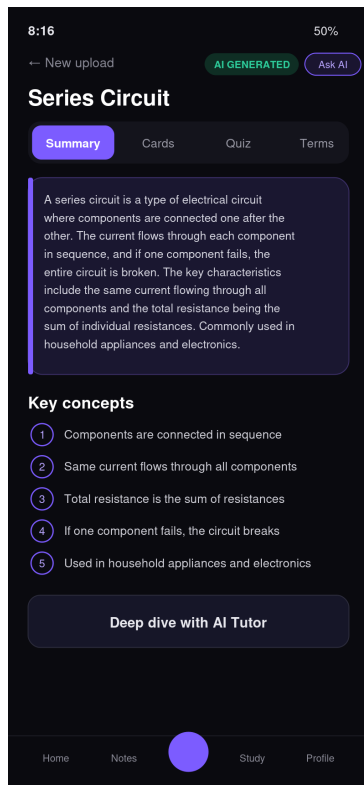


Figure A.2: AI-generated summary screen (left) and quiz screen (right)

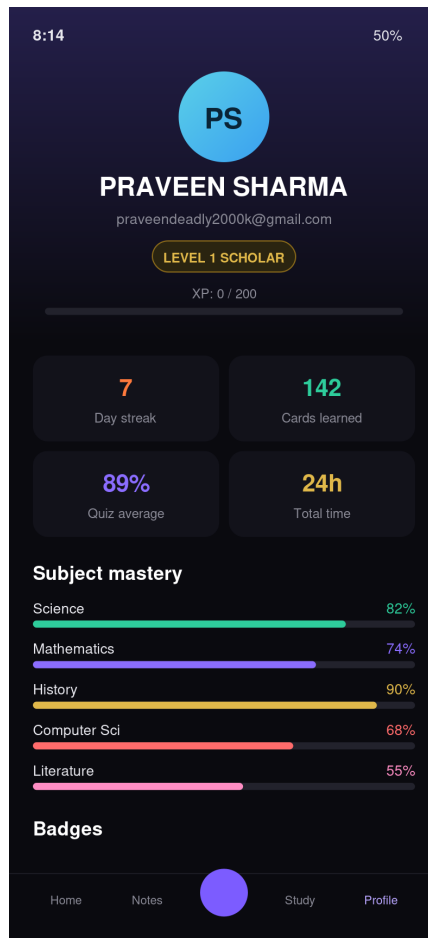


Figure A.3: Learner profile screen showing gamification and subject mastery

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